Data Cleaning Report

AeroGrid | Source: live Neon warehouse | UTC: 2026-11-13T21:11:41.876426+00:00

Run details

Latest ETL run ID: 11, mode=weekly, started=2026-11-13 20:34:19.325115+00:00, status=SUCCESS, version=wp11-v1

Staging and warehouse counts

Component	Stored records
Staging ADS-B positions (all statuses)	27479
Accepted staging positions (reject_reason IS NULL)	27479
Rejected staging positions (reject_reason set)	0
Warehouse tracking facts	27479
Stored rejection records	578
Staged FAA airspace features	252
Staged FAA restriction snapshots	0

Source coverage (loaded vs not loaded)

Source	Staging table	Accepted rows	Status
FAA restrictions	stg_restriction_snapshot	0	NO DATA LOADED
OpenSky (optional, D-013)	stg_opensky_state	0	NO DATA LOADED
AWC/IEM METAR	stg_weather_metar	400	LOADED
FAA airspace zones	stg_airspace_zone	252	LOADED
Nightjar / nominal-io DJI	stg_uas_position	22602	LOADED
Tartan ADS-B (KBTP)	stg_adsb_position	27479	LOADED

Coverage meaning

NO DATA LOADED means the source was validated but never ingested. It does not mean zero events occurred. OpenSky is optional under D-013 and has no ingestion path. See docs/13_accepted_data_limitations.md.

ETL run-level reconciliation (latest 12 nonzero runs)

ETL run	Mode	Read	Accepted	Rejected	Duplicates	Balanced?
11	weekly	23414	23254	160	0	YES
6	initial	27479	27479	0	0	YES
4	validate	27688	0	209	27479	YES
3	initial	27688	27479	209	0	YES

Reconciliation scope

For each ETL run shown: read = accepted + rejected + duplicates. These are per-run ledger counts, not a global total. The 418 stored stg_reject rows may span different runs or batches; do NOT add them to 27,479 staged positions to claim a source input total. Warehouse tracking rows represent accepted source-linked facts.

Source inventory and provenance (latest 25)

Source	License	Fetched UTC	Bytes	SHA-256
FAA-Airspace	U.S. government open data	2026-11-13 20:34:19.443551+00:00	1818975	e5ece4e131abc72f…
AWC	U.S. government open data (NOAA/NWS)	2026-11-13 20:34:19.443551+00:00	215184	c7e63f8095598c6b…
Nightjar	CC BY 4.0	2026-11-13 20:34:19.443551+00:00	9855064	6ef49014ee85104c…
Nominal-DJI	MIT (dataset card)	2026-11-13 20:34:19.443551+00:00	3010195	ba9cfc0783ff871c…
Nominal-DJI	MIT (dataset card)	2026-11-13 20:34:19.443551+00:00	4874707	b25b5618fc13506f…
Nominal-DJI	MIT (dataset card)	2026-11-13 20:34:19.443551+00:00	3606664	b64a8ea456da8d97…
FAA-Airspace	U.S. government open data	2026-11-13 20:34:19.443551+00:00	4473103	1baf6135dfa98d6c…
FAA-Airspace	U.S. government open data	2026-11-13 20:34:19.443551+00:00	45	7d09e532fc380630…
TARTAN	CC BY 4.0	2026-11-11 03:51:52.949066+00:00	4325376	86342d09e5004102…

Source URLs and attribution

Source 1: FAA-Airspace | URL: https://services6.arcgis.com/ssFJjBXIUyZDrSYZ/ArcGIS/rest/services/Route_Airspace/FeatureServer | Attribution: FAA UAS Data Delivery System ArcGIS services | Full SHA-256: e5ece4e131abc72f9ae7c3546af4562aacbfa81fb53ae03c00333f58deb98714 | Object key: None
Source 2: AWC | URL: https://aviationweather.gov/api/data/metar?ids=KBTP%2CKAGC&format=json&hours=168 | Attribution: NOAA/NWS Aviation Weather Center | Full SHA-256: c7e63f8095598c6bc31fd1ac18c11f7cadb962fdee4398ee996b6176fc070073 | Object key: None
Source 3: Nightjar | URL: https://huggingface.co/datasets/joshruby/nightjar-flight-20260804/resolve/main/dji/FlightRecord_2026-08-04_%5B12-46-25%5D.csv | Attribution: Nightjar flight telemetry dataset (joshruby), Hugging Face | Full SHA-256: 6ef49014ee85104c06f1b7fee743c71a0a06a4ba76b06e84fcde15f5091004d2 | Object key: None
Source 4: Nominal-DJI | URL: https://huggingface.co/datasets/nominal-io/dji-drone-telemetry/resolve/main/colorado-drone-data/FlightRecord_2024-08-02.csv | Attribution: nominal-io dji-drone-telemetry, Hugging Face | Full SHA-256: ba9cfc0783ff871c7099e243153dd8f23543cf531bf1dfe8ebaddc777c02e39d | Object key: None
Source 5: Nominal-DJI | URL: https://huggingface.co/datasets/nominal-io/dji-drone-telemetry/resolve/main/Palos%20Verdes/DJIFlightRecord_2025-02-17_%5B17-46-47%5D.csv | Attribution: nominal-io dji-drone-telemetry, Hugging Face | Full SHA-256: b25b5618fc13506f38f319cd6c779f6871539f2c14f71ab4e46d890f0db7492f | Object key: None
Source 6: Nominal-DJI | URL: https://huggingface.co/datasets/nominal-io/dji-drone-telemetry/resolve/main/drone_ocean/FlightRecord_2024-06-29.csv | Attribution: nominal-io dji-drone-telemetry, Hugging Face | Full SHA-256: b64a8ea456da8d97b42e4faa54b083345df799aab2614f478383e927c2cd8abf | Object key: None
Source 7: FAA-Airspace | URL: https://services6.arcgis.com/ssFJjBXIUyZDrSYZ/ArcGIS/rest/services/Class_Airspace/FeatureServer | Attribution: FAA UAS Data Delivery System ArcGIS services | Full SHA-256: 1baf6135dfa98d6c3114db46eb46168d54d76497b49265fc345cae0f797bfc69 | Object key: None
Source 8: FAA-Airspace | URL: https://services6.arcgis.com/ssFJjBXIUyZDrSYZ/ArcGIS/rest/services/Special_Use_Airspace/FeatureServer | Attribution: FAA UAS Data Delivery System ArcGIS services | Full SHA-256: 7d09e532fc380630caac6b4b1b5174a7a9f7a308f6544bcb79eefb97b1e12306 | Object key: None
Source 9: TARTAN | URL: https://airlab-cloud.andrew.cmu.edu:8080/swift/v1/AUTH_ac8533a83cff4d48bc8c608ad222d330/tartanaviation-adsb/kbtp/raw/2022.zip | Attribution: Patrikar et al., Scientific Data 12:468 (2025), DOI 10.1038/s41597-025-04775-6; TartanAviation | Full SHA-256: 86342d09e5004102d765b82d04d16b43b6f15f2e7d829bfbde7909187d08a762 | Object key: samples/kbtp/01-01-22/1.csv

Rejected records by source and rule

Source	Reason	Count
TARTAN	INVALID_VEHICLE_ID	416
NOMINAL_DJI	ZERO_COORDINATE	159
TARTAN	INVALID_TIMESTAMP	2
NIGHTJAR	INVALID_TIMESTAMP	1

Cleaning methods

Source normalizers validate timestamps, position units, identifiers, and coordinates. Source rows are tracked using staging reject_reason, rejection records and ETL run logs. Accepted staging counts come from reject_reason IS NULL, while separate rejected records may be recorded in stg_reject. Never assume their totals share the same source-run scope. Source-specific duplicates are recorded by ETL run.

Limitations and provenance

January 2022 KBTP Tartan positions are historical. A live FAA TFR service was checked in November 2026; it returned zero features in the selected study box at that time. No FAA restriction snapshots are stored in the warehouse for this sample. Do not join current restrictions to historical flights. Original on_ground values remain NULL; airborne eligibility is inferred under WP07-AIR-01. The physical spatial grid is 2D (0.01 degree), not an altitude-band grid.

Data authenticity

Production evidence is based on source-linked real rows, not synthetic demonstration records. Zero-source coverage is reported as zero, not imputed.

Reproduction

Run WP-04 staging checks, WP-06 core fact reconciliation and WP-07 derived facts verifier through protected warehouse CI. Regenerate this PDF using python -m etl.reports with the read-only Neon connection in CI.